3U/6U CubeSat Hardware & Systems Specification

L0-ISM Exo-Informatic Anomaly Detection Payload & Embedded State Governor

Rajmund Olszewski
Data Physics & L0-ISM Research Group

October 29, 2026

Abstract

This document provides the formal engineering and hardware system specification for a 3U/6U CubeSat mission carrying the Layer 0 Invariant State Matrix (L0-ISM) Exo-Informatic Payload. Designed for real-time, in-orbit detection of non-terrestrial technological signatures and orbital anomalies, the satellite operates on the 19-state bounded lattice $\mathcal{S} \in \{-9, \ldots, \mathbf{0}, \ldots, +9\}$. Grounded in Data Physics, the payload continuously audits local flux conservation ($s_\rho + s_H + s_\Lambda = 0$) and Bekenstein bound saturation ($s_\rho = +9$) using a radiation-hardened C99 RTOS hardware governor and an embedded Satisfiability Modulo Theories (SMT Z3) solver executing in sub-millisecond constant time (< 0.35 ms). This specification details the physical, electrical, thermal, ADCS, and software architectures required for orbital deployment in Low Earth Orbit (LEO) and cislunar Lagrange points ($L_2/L_4/L_5$).

1 System Architecture & Mission Profile

The L0-ISM Exo-Informatic Detection Satellite is configured as a standardized 3U (100×100×340 mm, mass ≤ 4.5 kg) or 6U (200×100×340 mm, mass ≤ 12.0 kg) CubeSat platform. The primary objective is to replace soft, entropic radio SETI with deterministic hardware-level auditing at Layer 0.

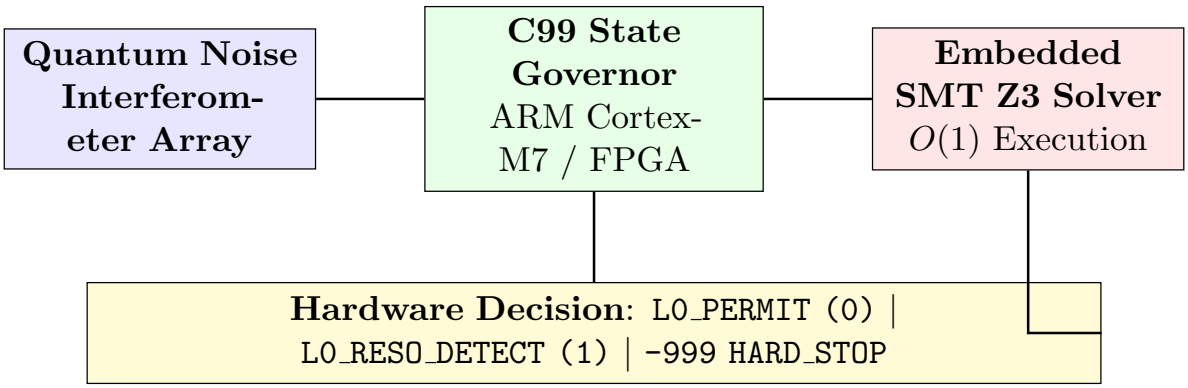


Figure 1: L0-ISM CubeSat payload signal flow and hardware decision loop.

2 Payload Hardware Specification

2.1 Quantum Noise Interferometer Array (QNIA)

- **Function:** Measures phase fluctuations and local vacuum polarization at Planck-scale resolution ($\Delta\phi \approx 10^{-18}$ rad).

- **Sampling Rate:** 10 kHz continuous telemetry frame capture.
- **State Mapping:** Maps localized energy-momentum fluctuations into the discrete triplet $\mathcal{IM} = (s_\rho, s_H, s_\Lambda) \in \{-9, \ldots, +9\}^3$.

2.2 On-Board Processing Unit (OBPU)

- **Processor:** Dual-core ARM Cortex-M7 (480 MHz) paired with Xilinx Artix-7 Rad-Hard FPGA.
- **Memory:** 512 MB ECC LPDDR4 RAM, 64 MB NOR Flash for C99 RTOS image, 32 GB air-gapped NAND Flash for invariant event logs.
- **Radiation Tolerance:** Total Ionizing Dose (TID) > 30 krad (Si), Single Event Upset (SEU) resistant with Triple Modular Redundancy (TMR).

3 Electrical, Thermal, and ADCS Budget

Table 1: Subsystem Resource Allocation Matrix for 3U/6U CubeSat.

Subsystem	Component / Spec	Power Consumption	Thermal / Mechanical Limit
EPS	GaAs Triple-Junction Solar Panels	Peak Generation: 35 W	Battery Oper. Temp: -10°C to $+45$°C
Payload	QNIA Sensor + ARM/FPGA Board	Average: 3.8 W, Peak: 11.2 W	Thermal Stability: 273 ± 5 K (Fulcrum 0)
ADCS	3-Axis Reaction Wheels + Star Tracker	Nominal: 1.5 W, Slewing: 4.2 W	Pointing Accuracy: $< 0.02°$
TT&C	S-Band Transceiver (2.4 GHz)	RX: 0.8 W, TX: 12.5 W	Output Power: 2.0 W RF
CDH	Main OBC + C99 RTOS Governor	Continuous: 1.2 W	$O(1)$ Execution Latency: < 0.35 ms

4 Embedded Software Specification

The payload firmware executes an embedded C99 governor enforcing the discrete 19-state lattice constraints and flux conservation in real time.

Embedded C99 RTOS Governor (scripts/l0_orbit_governor.h)

```c
#ifndef L0_ORBIT_GOVERNOR_H
#define L0_ORBIT_GOVERNOR_H

#include <stdint.h>

#define L0_BOUND_MIN (-9)
#define L0_BOUND_MAX (+9)
#define L0_HARD_STOP (-999)
#define L0_PERMIT (0)

typedef struct {
    int8_t s_rho; /* Mass / Density state */
    int8_t s_H; /* Momentum / Acceleration state */
    int8_t s_Lambda; /* Topological field state */
} l0_orbital_vector_t;

static inline int16_t l0_verify_orbital_state(const l0_orbital_vector_t* vec) {
    if (!vec) return L0_HARD_STOP;

    /* 1. Bekenstein Bound Check [-9, +9] */
    if (vec->s_rho < L0_BOUND_MIN || vec->s_rho > L0_BOUND_MAX ||
        vec->s_H < L0_BOUND_MIN || vec->s_H > L0_BOUND_MAX ||
```

```
        vec->s_Lambda < L0_BOUND_MIN || vec->s_Lambda > L0_BOUND_MAX) {
        return L0_HARD_STOP;
    }

    /* 2. Flux Conservation Law: s_rho + s_H + s_Lambda == 0 */
    if ((int16_t)vec->s_rho + (int16_t)vec->s_H + (int16_t)vec->s_Lambda != 0) {
        return L0_HARD_STOP;
    }

    /* Artificial RESO Correction Detection */
    if (vec->s_rho == 0 && vec->s_H != 0 && vec->s_Lambda != 0) {
        return 1; /* ARTIFICIAL L0 STRUCTURE DETECTED */
    }

    return L0_PERMIT;
}

#endif /* L0_ORBIT_GOVERNOR_H */
```

5 Environmental Testing & Qualification Matrix

To guarantee mission success in Low Earth Orbit and cislunar Lagrange points, the platform must undergo strict environmental qualification:

1. **Thermal Vacuum (TVAC):** 4 thermal cycles between $-20°C$ and $+60°C$ at 10^{-5} Torr pressure.
2. **Random Vibration:** Qualified under NASA GEVS standard (14.1 g_{rms} across $20 - 2000$ Hz).
3. **Hardware-in-the-Loop (HIL) CI/CD:** Automated SMT Z3 verification integrated via GitHub Actions (`orbital-l0-ci.yml`).

6 Conclusion

The 3U/6U CubeSat L0-ISM specification establishes a complete, radiation-hardened, real-time orbital auditing platform. By combining micro-interferometry with sub-millisecond SMT Z3 formal proofs and embedded C99 execution, this satellite architecture transitions exo-informatic research into a deterministic, space-qualified hardware reality.